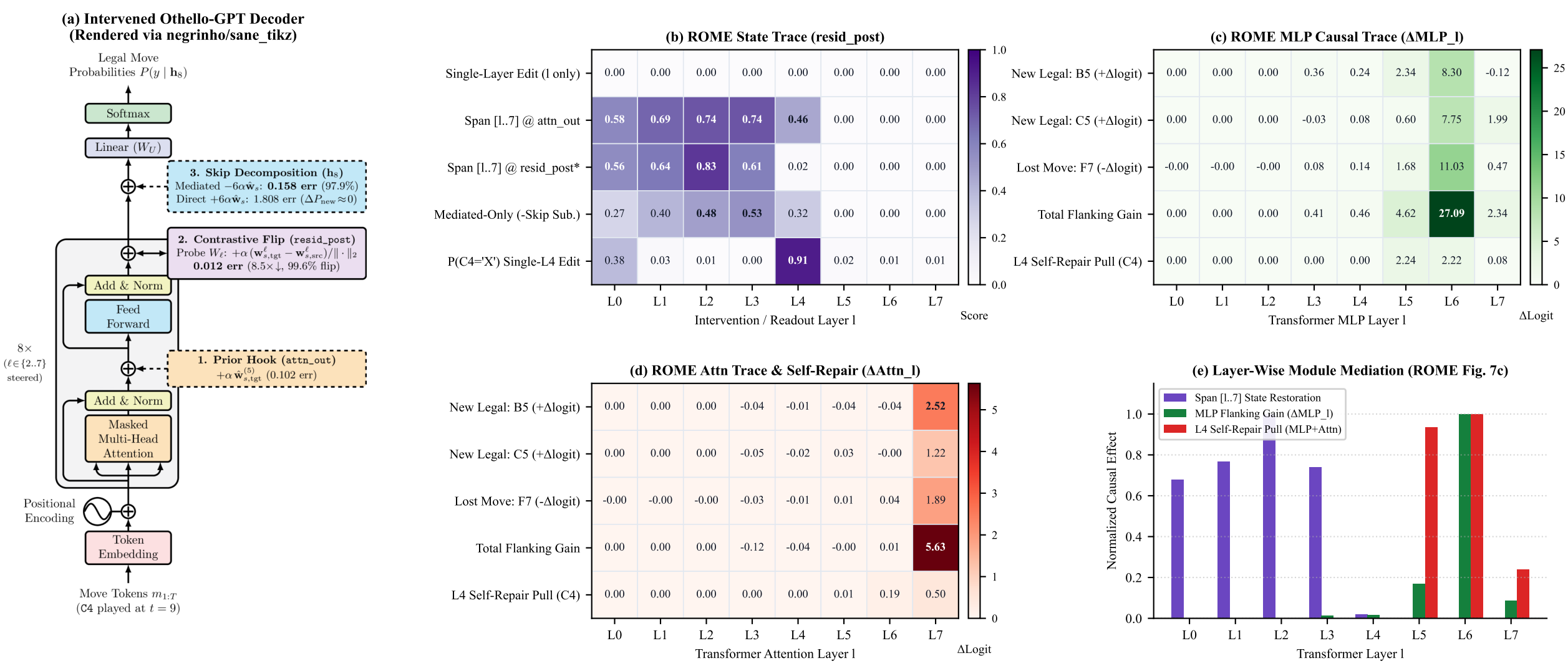

(a) Intervened Othello-GPT Decoder
(Rendered via negrinho/sane_tikz)
Legal Move
Probabilities $P(y \mid \mathbf{h}_8)$
Softmax
Linear ($W_U$)
3. Skip Decomposition ($\mathbf{h}_8$)
Mediated $-6\alpha\hat{\mathbf{w}}_s$: 0.158 err (97.9%)
Direct $+6\alpha\hat{\mathbf{w}}_s$: 1.808 err ($\Delta P_{\text{new}} \approx 0$)
2. Contrastive Flip (resid_post)
Probe $W_\ell$: $+\alpha\,(\mathbf{w}^{\ell}_{s,\text{tgt}} - \mathbf{w}^{\ell}_{s,\text{src}})/\|\cdot\|_2$
0.012 err (8.5×↓, 99.6% flip)
Add & Norm
Feed Forward
8×
($\ell \in \{2..7\}$ steered)
1. Prior Hook (attn_out)
$+\alpha\,\hat{\mathbf{w}}^{(5)}_{s,\text{tgt}}$ (0.102 err)
Add & Norm
Masked Multi-Head Attention
Positional Encoding
Token Embedding
Move Tokens $m_{1:T}$
(C4 played at $t = 9$)
(b) ROME State Trace (resid_post)
Single-Layer Edit (l only)
Span [l..7] @ attn_out
Span [l..7] @ resid_post*
Mediated-Only (-Skip Sub.)
P(C4='X') Single-L4 Edit
0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
0.58 0.69 0.74 0.74 0.46 0.00 0.00 0.00
0.56 0.64 0.83 0.61 0.02 0.00 0.00 0.00
0.27 0.40 0.48 0.53 0.32 0.00 0.00 0.00
0.38 0.03 0.01 0.00 0.91 0.02 0.01 0.01
L0 L1 L2 L3 L4 L5 L6 L7
Intervention / Readout Layer l
Score
(c) ROME MLP Causal Trace (ΔMLP_l)
New Legal: B5 (+Δlogit)
New Legal: C5 (+Δlogit)
Lost Move: F7 (-Δlogit)
Total Flanking Gain
L4 Self-Repair Pull (C4)
0.00 0.00 0.00 0.36 0.24 2.34 8.30 -0.12
0.00 0.00 0.00 -0.03 0.08 0.60 7.75 1.99
-0.00 -0.00 -0.00 0.08 0.14 1.68 11.03 0.47
0.00 0.00 0.00 0.41 0.46 4.62 27.09 2.34
0.00 0.00 0.00 0.00 0.00 2.24 2.22 0.08
L0 L1 L2 L3 L4 L5 L6 L7
Transformer MLP Layer l
ΔLogit
(d) ROME Attn Trace & Self-Repair (ΔAttn_l)
New Legal: B5 (+Δlogit)
New Legal: C5 (+Δlogit)
Lost Move: F7 (-Δlogit)
Total Flanking Gain
L4 Self-Repair Pull (C4)
0.00 0.00 0.00 -0.04 -0.01 -0.04 -0.04 2.52
0.00 0.00 0.00 -0.05 -0.02 0.03 -0.00 1.22
-0.00 -0.00 -0.00 -0.03 -0.01 0.01 0.04 1.89
0.00 0.00 0.00 -0.12 -0.04 -0.00 0.01 5.63
0.00 0.00 0.00 0.00 0.00 0.01 0.19 0.50
L0 L1 L2 L3 L4 L5 L6 L7
Transformer Attention Layer l
ΔLogit
(e) Layer-Wise Module Mediation (ROME Fig. 7c)
Span [l..7] State Restoration
MLP Flanking Gain (ΔMLP_l)
L4 Self-Repair Pull (MLP+Attn)
Normalized Causal Effect
L0 L1 L2 L3 L4 L5 L6 L7
Transformer Layer l